

Requirements: Project Timing from Standard Blocks

TOC-PLM-07 · status 10/02/2026 · internal working document, not controlled

Purpose / Scope:

Project timelines are built today by copying tasks in MS Project. This specification defines how PLM2 and the SEP project database generate the timing for project members from standard blocks instead: the standard steps are defined once, the project only enters its molds, parts, team and anchor dates, and only project-specific work is planned by hand. It records what the analysis of the G6x timeline found, the requirements with their acceptance test, what PLM2 already has, and a traceability matrix from every requirement to its implementation.

Owner / involved:

- C. Demmler, Process Development: owner of PLM2 and of this specification, keeps the trace current
- Project Manager of project 1748 (author of the G6x timeline): reviews the blocks, pilot user
- Program Management: users of the project timing, sign off the requirements
- Quality / APQP: PPAP and capability steps, SEP gate content
- Process Engineering, Metrology, Packaging, Toolshop, Production Planning: owners of the steps in the blocks

1. Starting point: the G6x timeline (analysed 10/02/2026)

File: G6x Project Timeline (2026-09-30).mpp, MS Project, 239 tasks, 29 resources, 04/06 to 11/11/2026. It covers the G45 WHL transfer, three mold engineering changes (3360, 3349, 3369) and ten G67 mold launches.

Pattern	Rows	What it really is
G67 mold launch: 9 molds x the same 11 steps (+1 partial)	~104	One standard block, applied 9 times
Cpk split per characteristic (weight, thickness, SPC 1-6)	24	A checklist on one task
iPQ elements 0.1 to 5.6, identical dates and owner	15	The PPAP / iPQ element checklist
ISIR documents (CP, WI, gauge WI, part history, cover sheet)	~8	Same checklist
Mold changes 3360 / 3349 / 3369	~55	Engineering changes (ECR), planned in the project file
Really project-specific (BMW vehicle trial, Baxter payment, Hall 52 downtime...)	~30	The only part that needs hand planning

1.1 Findings

No.	Found in the file	Consequence
TF-01	0 of 239 tasks linked, no constraints, no baseline	Every date typed by hand; a slip moves nothing and cannot be measured
TF-02	Molds 3362, 3343, 3352: unpack (08/31 to 09/03) before delivery (09/03)	Impossible order, not visible
TF-03	Mold 3376: no TH1; TH2 on the day of gauge validation	Missing standard step
TF-04	Mold 3362: TH1 on the day of gauge validation	Dependency not respected
TF-05	Mold 3377: press 1300-2 in some lines, 1300-3 in others	Copy error
TF-06	Task 5.9.11 named "TH2" but is the 30-pc Cpk	Copy error
TF-07	Same person as 3 resources (holmes / Christopher.holmes / christopher.holmes), Rautzenberg 3x, a resource named "Mold 3345: Verify correct AI Level..."	Workload per person not usable
TF-08	Cpk and iPQ elements as single rows (39 rows)	Plan unreadable for the team
TF-09	Mold changes planned as project tasks (~55 rows)	Duplicates the ECR process in PLM2
TF-10	One 239-line Gantt for everybody	Team members cannot see their own next tasks

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1.2 Pilot (built 10/02/2026)

In docs/project-timing/: **G6x Timing Template.xlsx** (block library, team, inputs, formula-driven plan, findings), **G67 Timeline (from template).xml** (MS Project, 99 tasks, 72 links, 14 resources) and **generate.py** (rebuilds both). Per mold only press, part numbers, delivery, gauge validation date and TH1 trial slot are entered. All 99 Excel dates were checked against the generator: 0 differences. Against the PM's hand-typed dates 38 of 53 are identical; the 15 others are TF-02, TF-03 and TF-04 and hand-picked unpack days.

2. Target concept

Layer	Content	Where
What (per gate)	SEP matrix GB-DP-0001: 7 gates, 232 work items, PM + Quality sign-off	SEP project database (in place)
How and when (per mold / part)	Standard blocks: steps, roles, durations, links, checklists	PLM2 block library (new)
Project	Molds and parts from the PDB, team (role -> person), anchor dates, project-specific tasks	PLM2 project (partly in place)
Changes	Mold engineering changes as ECR with their own plan; the project shows the milestones	PLM2 ECR (in place)

Principle: nobody copies a task. A standard step exists once in a block; a project gets it by applying the block to its molds or parts. Improving a block improves every new project.

3. What PLM2 already has

Component	Today	Gap for project timing
Gantt 2.0 engine (plan_engine)	Links FS/SS/FF/SF with lag, constraints, working calendar, critical path, baseline, progress, idea blocks (bank build)	Runs on change plans only, not on project plans
SEP project database	Template copied per project: 7 gates, 232 items with department, responsible, status, audit trail; gate target date linked to a milestone	Items have no dates, durations or links; no per-mold breakdown
Project milestones	Name, due date, status per project	No link to tasks
Project team (project_responsibles)	One responsible per role per project, main / backup counting	Roles for Metrology, CMM lab, trial lead, plant layout, toolshop, production planning missing
Items (parts)	Tools, articles, gauges; tool produces article, gauge checks article; planned press (tool_machine), cavities, toolmaker	Tool delivery date not held
PPAP submission	Per part revision: level, customer, 18 elements with status and file	Only the AIAG element set; VDA iPQ set missing
ECR change plan	Stages 4-8 timing per change with the same engine	No standard plan template
Control plan characteristics	Not in PLM2	Needed for the Cpk checklist

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4. Requirements

4.1 Block library

No.	Requirement	Acceptance (how we test it)
TR-01	A block is master data: steps with ID, name, role, duration in workdays, dependency (FS/SS/FF, lag), mandatory/optional, checklist and SEP item reference. Blocks are created and edited in PLM2 without code changes.	The three pilot blocks LAUNCH, MOLDCHG, PPAP are entered in PLM2 and match the Excel pilot
TR-02	Each block has an owner, a version and a status (draft / released). Changes are audited. A plan keeps the block version it was generated from; a new version reaches a running plan only when the PM applies it.	Release v2 of LAUNCH: existing G67 plan unchanged until applied
TR-03	Steps name roles, never persons. The role list is the project-team role list, extended by the roles the blocks need (decision D-01).	No person is stored in any block
TR-04	A step carries a checklist instead of extra rows. A checklist can be fixed text or come from data: control plan characteristics (TR-16), customer PPAP element set (TR-15).	G67 Cpk task shows weight, thickness, SPC 1-n of that part; 0 rows per characteristic

4.2 Generating the plan from the PDB

No.	Requirement	Acceptance (how we test it)
TR-05	A block is applied to items of a project: LAUNCH per tool, PPAP per part family, MOLDCHG per ECR. Task names come from the block and the item (press from the planned press, part numbers from the produces relation).	Select the 9 G67 tools -> 99 tasks, names as in the pilot
TR-06	Per item only anchor dates are entered (e.g. delivery, gauge validation, trial slot). All other dates are calculated on the plant's working calendar with holidays.	G67 dates equal the Excel pilot (99 of 99)
TR-07	Project-specific tasks are added by hand next to the generated ones and can be linked to them.	BMW vehicle trial linked after TH2 of mold 3349 moves with it
TR-08	Items can be added or dropped later: a new tool gets its block without touching the others; a dropped one is cancelled with a reason, not deleted.	Add a 10th tool to G67: 11 new tasks, others unchanged

4.3 Scheduling and tracking

No.	Requirement	Acceptance (how we test it)
TR-09	Project plans use the Gantt 2.0 engine: links, constraints, critical path. A moved date moves its successors; nobody types successor dates.	Delay one delivery by 5 workdays: every successor moves 5 workdays
TR-10	Logic checks warn about impossible order (unpack before delivery, trial before gauge) and missing mandatory steps; dropping a mandatory step needs a reason.	TF-02, TF-03, TF-04 are flagged when the G6x data is entered
TR-11	A baseline is set when the plan is released. Progress and actual dates are recorded per task; slip against the baseline is visible per task and item.	Baseline + one late task shows its slip in workdays
TR-12	After the baseline, moving a date asks for a reason; the change is audited (who, when, old, new, reason).	Audit list shows the move with reason

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4.4 Integration with SEP, ECR and PPAP

No.	Requirement	Acceptance (how we test it)
TR-13	Each block step references one or more SEP work items (proposal in section 5). The SEP item shows the progress of its tasks over all items of the project; when all are done, closing the item is proposed to its responsible.	SEP item "Assessment of short-term capabilities CP and CPK" shows 9 of 9 Cpk tasks
TR-14	SEP gate target dates are checked against the block milestones; a block milestone after its gate target is flagged.	Move gate B/RG6 before the last Cpk: warning
TR-15	Mold engineering changes run as ECR; MOLDCHG is the standard template of the change plan. The project timing shows only the ECR milestones, read-only, linked to the change.	G6x sections 2-4 = 3 ECRs and 0 copied rows in the project plan
TR-16	Control plan characteristics per article are held in the PDB (characteristic, type: weight / thickness / SPC, special characteristic flag) and feed the Cpk checklists.	Article 5A65DF8 shows its characteristics; its Cpk task lists them
TR-17	The PPAP block's checklist is the PPAP submission's element list; the element set follows the customer (AIAG 18 elements or VDA iPQ).	BMW part shows the iPQ list of the G45 file

4.5 Views and output

No.	Requirement	Acceptance (how we test it)
TR-18	"My tasks" per person: tasks of the roles they hold on each project team, next 2-3 weeks, with main / backup counting as for ECR tasks.	Metrology sees only gauge tasks, by date
TR-19	PM view: Gantt per project, filters by item, role, block; exception list (late, at risk, missing person for a role).	G67 filtered to mold 3355 shows its 11 steps
TR-20	Exchange with MS Project: export as MS Project XML with links and resources (as the pilot).	Export opens in MS Project with links intact
TR-21	KPIs: milestones on time (e.g. TH1, mold ready for PPAP, PPAP approval) with the calendar-day rule of the ECR KPI board.	KPI board shows on-time rate per milestone type

4.6 Rights

No.	Requirement	Acceptance (how we test it)
TR-22	The project PM edits the project plan; block library editing only by block owners and admins; team members report progress on their own tasks.	Team member cannot move another role's task

5. Block steps against SEP work items (proposal, to confirm in the workshop)

Step	Block step	SEP work item (gate, no.)
L01	Assign location in plant layout	Determine space requirements including layout (K/RG2 31, D/RG4 10, D/RG4 14)
L04	Gauge validation	Create and release specifications (gauges) (E/RG3 25)
L06	Complete work instructions	Create / revise test and work instructions (K/RG2 24, E/RG3 26)
L08	Packaging trial / instruction	Determining and implementing the packaging and logistics concept (K/RG2 33)
L09	6-pc dimensional (CMM)	Create measurement orders, evaluate measurement reports (C/RG5 17)
L10	30-pc Cpk	Start capabilities (C/RG5 15); assessment of short-term capabilities CP and CPK (B/RG6 18)
L11	Mold ready for PPAP	Start handover of series tools to the production plant (B/RG6 27)
P02	ISIR / iPQ package	Initial sampling customer grade 3 (B/RG6 21); grade 1 (A/RG7 6)
P04	Full PPAP approval	Assessment of long-term capabilities (A/RG7 7)

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Step	Block step	SEP work item (gate, no.)
E01-E13	Mold engineering change	Change management items (Sales, every gate) -> runs as ECR

6. Steps and timing (proposal)

No.	Step	Who	Due
S1	Pilot: use the Excel / MS Project template for the open G67 launches	C. Demmler, PM 1748	10/16/2026
S2	Block workshop: steps, durations, roles, SEP mapping (section 5), decisions D-01 to D-05	PM 1748, Quality, Process Eng.	10/30/2026
S3	Review and sign-off of this specification	Program Mgmt., Quality	11/13/2026
S4	Phase 1: block library, generation, engine on project plans, baseline (TR-01..12, TR-22)	C. Demmler	02/26/2027
S5	Phase 2: SEP link, ECR milestones, control plan characteristics, PPAP element sets, my tasks (TR-13..18)	C. Demmler	04/30/2027
S6	Phase 3: PM view and exceptions, MS Project export, KPIs (TR-19..21)	C. Demmler	06/25/2027
S7	Acceptance: rebuild G67 in PLM2 and compare with the pilot; first new project planned only in PLM2	PM 1748, C. Demmler	07/30/2027

Phase 1 starts after the ECR go-live on 12/21/2026, so the ECR rollout is not delayed.

7. Open decisions

No.	Decision	Who
D-01	Role list: extend the project team by Metrology, CMM lab, trial lead, plant layout, toolshop, production planning, or map them to existing roles	Program Mgmt., Quality
D-02	Which block steps are mandatory (cannot be dropped without a reason)	Workshop S2
D-03	SEP item closing: proposed to the responsible (default) or automatic	Quality
D-04	Owner of the block library: Program Management or Process Development	Program Mgmt.
D-05	MS Project after go-live: export only, or also import	PM 1748

8. Traceability matrix (status 10/02/2026)

Every requirement is traced to where it comes from, what PLM2 already has and the evidence of its implementation. Every implementation commit names its TR number; the living copy is docs/project-timing/TRACE.md and this table is updated from it at each revision.

No.	From	PLM2 basis	Status	Evidence
TR-01	TF-08, pilot	-	Pilot	Blocks sheet, generate.py
TR-02	Concept	Audit pattern of SEP items	Open	
TR-03	TF-07	project_responsibles	Partial	Team sheet; D-01 open
TR-04	TF-08	-	Pilot	Checklist column (text)
TR-05	Pilot	Items, produces relation, tool_machine	Pilot	Plan sheet, 99 tasks
TR-06	TF-01	plan_engine calendar	Pilot	99 of 99 dates verified 10/02
TR-07	Section 1	Change plan free tasks	Open	
TR-08	Concept	-	Open	
TR-09	TF-01	plan_engine (change plans)	Partial	MS Project XML: 72 links
TR-10	TF-02..04	plan_engine validation warnings	Partial	Pilot plans TH1 after gauge
TR-11	TF-01	plan_engine baseline, progress	Partial	

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No.	From	PLM2 basis	Status	Evidence
TR-12	Concept	Release deadline move with reason (F-02)	Partial	
TR-13	Concept	SEP work items	Partial	Mapping proposal, section 5
TR-14	Concept	SEP gate target date	Partial	
TR-15	TF-09	ECR change plan	Partial	MOLDCHG block defined
TR-16	TF-08	-	Open	
TR-17	TF-08	PPAP submission (AIAG)	Partial	iPQ list in PPAP block
TR-18	TF-10	My Tasks (ECR), main/backup	Partial	
TR-19	TF-10	Gantt 2.0 UI (change plans)	Partial	
TR-20	Pilot	-	Pilot	G67 Timeline (from template).xml
TR-21	Concept	ECR KPI board	Partial	
TR-22	Concept	Project team rights	Partial	

Status: Open = nothing yet; Partial = PLM2 has the basis, the timing function is missing; Pilot = proven in the Excel / MS Project pilot, not in PLM2; Done = in PLM2 and the acceptance test passed.